
Physics 523 HW 3

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I use the notation in my Real Scalar Field Notes. I usually use p for the magnitude of the momentum 3-vector, $p = |\mathbf{p}|$, so $p^2 = \mathbf{p} \cdot \mathbf{p}$. But for this homework I will use p^2 for the dot product of the momentum 4-vector: $p^2 = \vec{p} \cdot \vec{p} = \vec{p}^2 = (p^0)^2 - \mathbf{p}^2$.

Part I

1

1. LAGRANGIAN

The bare (B) Lagrangian is

$$\mathcal{L} = -\frac{\Phi_B \square \Phi_B}{2} - \frac{m_B^2 \Phi_B^2}{2} - \frac{\lambda_B}{4!} \Phi_B^4. \quad (1)$$

I introduce renormalization factors

$$\Phi_B = \sqrt{Z_\Phi} \Phi, \quad m_B = \sqrt{Z_m} m, \quad \lambda_B = Z_\lambda \lambda. \quad (2)$$

$$\mathcal{L} = -\frac{Z_\Phi}{2} \Phi \square \Phi - Z_m Z_\Phi \frac{m^2}{2} \Phi^2 - Z_\lambda Z_\Phi^2 \frac{\lambda}{4!} \Phi^4 \quad (3)$$

$$Z_\Phi = 1 + \delta_\Phi, \quad Z_m = 1 + \delta_m, \quad Z_\lambda = 1 + \delta_\lambda \quad (4)$$

$$\begin{aligned} \mathcal{L} = & -\frac{\Phi \square \Phi}{2} - \frac{\delta_\Phi}{2} \Phi \square \Phi - (1 + \delta_\Phi + \delta_m + \delta_m \delta_\Phi) \frac{m^2}{2} \Phi^2 \\ & - (1 + 2\delta_\Phi + \delta_\lambda + \delta_\Phi^2 + 2\delta_\lambda \delta_\Phi + \delta_\lambda \delta_\Phi^2) \frac{\lambda}{4!} \Phi^4 \end{aligned} \quad (5)$$

$$\begin{aligned} \mathcal{L} = & -\frac{\Phi \square \Phi}{2} - \frac{m^2}{2} \Phi^2 - \frac{\lambda}{4!} \Phi^4 - \frac{\delta_\Phi}{2} \Phi \square \Phi \\ & - (\delta_\Phi + \delta_m + \delta_m \delta_\Phi) \frac{m^2}{2} \Phi^2 \\ & - (2\delta_\Phi + \delta_\lambda + \delta_\Phi^2 + 2\delta_\lambda \delta_\Phi + \delta_\lambda \delta_\Phi^2) \frac{\lambda}{4!} \Phi^4 \end{aligned} \quad (6)$$

The free and interaction Lagrangians are

$$\boxed{2\mathcal{L}_0 = -\Phi(\square + m^2)\Phi}, \quad \boxed{\mathcal{L}_1 = -\frac{\lambda}{4!} \Phi^4}. \quad (7)$$

The counterterm (C) Lagrangian is

$$\begin{aligned} \mathcal{L}_C = & -\frac{1}{2} \Phi [\delta_\Phi \square + (\delta_\Phi + \delta_m + \delta_m \delta_\Phi) m^2] \Phi \\ & - (2\delta_\Phi + \delta_\lambda + \delta_\Phi^2 + 2\delta_\lambda \delta_\Phi + \delta_\lambda \delta_\Phi^2) \frac{\lambda}{4!} \Phi^4. \end{aligned} \quad (8)$$

$$\boxed{\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_C} \quad (9)$$

2. ONE LOOP LAGRANGIAN

I expand the counterterm Lagrangian in the renormalized coupling λ . The counterterms are first order: $\delta \sim \mathcal{O}(\lambda)$,

$$\mathcal{L}_C = -\frac{1}{2} \Phi [\delta_\Phi \square + (\delta_\Phi + \delta_m) m^2] \Phi. \quad (10)$$

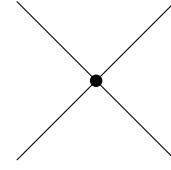


FIG. 1. Interaction vertex with factor $-i\lambda$.

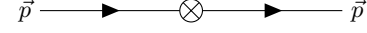


FIG. 2. Counterterm interaction vertex with factor $i[p^2 \delta_\Phi - (\delta_\Phi + \delta_m) m^2]$, see Schwartz Eq. 19.13.

But δ_λ has dropped out. I could go to second order (λ^2) but that would be overkill. I want all δ to leading order: so I expand in loops, not λ . In this case expanding to one loop order means expanding to $\mathcal{O}(\delta)$.

$$\boxed{\mathcal{L}_C = -\frac{1}{2} \Phi [\delta_\Phi \square + (\delta_\Phi + \delta_m) m^2] \Phi - (2\delta_\Phi + \delta_\lambda) \frac{\lambda}{4!} \Phi^4} \quad (11)$$

Plugging this into Eq. 9 yields the one loop Lagrangian.

3. FEYNMAN RULES

The propagator is

$$\Phi \sim \frac{i}{p^2 - m^2 + i\epsilon}. \quad (12)$$

I show the interaction vertex in Fig. 1, and the counterterm interaction vertices in Figs. 2 & 3.

4. DIMREG

In 4D

$$[\Phi] = 1, \quad [\mathcal{L}] = 4. \quad (13)$$

$$4 = [\mathcal{L}] = [\lambda \Phi^4] = [\lambda] + 4 \implies [\lambda] = 0 \quad (14)$$

When using DimReg we want to keep the coupling dimensionless in d dimensions, so as in QED we introduce a factor: $\lambda \rightarrow \mu^{4-d} \lambda$.

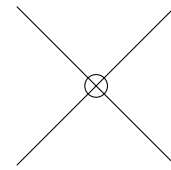
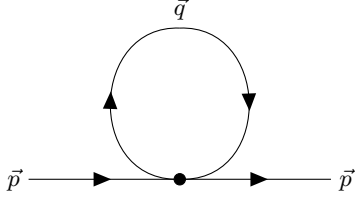


FIG. 3. Counterterm interaction vertex with factor $-i(2\delta_\Phi + \delta_\lambda) \lambda$.


 FIG. 4. Feynman diagram for $i\mathcal{G}^{(0)}(\vec{p})$.

 FIG. 5. $\mathfrak{G}$

5. PROPAGATOR

The exact propagator (2-point Green's function) is

$$\langle \Omega | T[\Phi(\vec{x})\Phi(\vec{y})] | \Omega \rangle = \frac{i}{(2\pi)^4} \int d^4p e^{i\vec{p} \cdot (\vec{y} - \vec{x})} \mathcal{G}(\vec{p}). \quad (15)$$

I want its momentum space representation $\mathcal{G}(\vec{p})$.

5.1. Lowest Order

The lowest order propagator (Fig. 4) is Eq. 12

$$i\mathcal{G}^{(0)}(\vec{p}) = \frac{i}{p^2 - m^2}, \quad (16)$$

where I have taken the $\epsilon \rightarrow 0$ limit. The factor of i comes from following Schwartz's (Sec. 18.2) normalization convention.

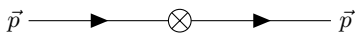
$$\boxed{\mathcal{G}^{(0)}(\vec{p}) = \frac{1}{p^2 - m^2}} \quad (17)$$

5.2. First Correction

$$\mathcal{G}^{(1)}(\vec{p}) = \mathcal{G}^{(0)}(\vec{p}) + \Delta\mathcal{G}(\vec{p}), \quad \Delta\mathcal{G} = \Delta\mathcal{G}^{(1)} \quad (18)$$

The leading (first) order correction (in λ) comes from the diagrams in Figs. 5 & 6.

$$i\Delta\mathcal{G}(\vec{p}) = \mathfrak{G} + \mathfrak{H} \quad (19)$$


 FIG. 6. $\mathfrak{H}$

$$\mathfrak{H} = \frac{i}{p^2 - m^2} i[p^2\delta_\Phi - (\delta_\Phi + \delta_m)m^2] \frac{i}{p^2 - m^2} \quad (20)$$

$$\mathfrak{H} = i \frac{(\delta_\Phi + \delta_m)m^2 - p^2\delta_\Phi}{(p^2 - m^2)^2} \quad (21)$$

$$i\Delta\mathcal{G}(\vec{p}) = \mathfrak{G} + i \frac{(\delta_\Phi + \delta_m)m^2 - p^2\delta_\Phi}{(p^2 - m^2)^2} \quad (22)$$

$$\mathfrak{G} = \frac{i}{p^2 - m^2} i\Sigma(\vec{p}) \frac{i}{p^2 - m^2} = -\frac{i\Sigma(\vec{p})}{(p^2 - m^2)^2} \quad (23)$$

Here $i\Sigma(\vec{p})$ is the amputated version of $\mathfrak{G}$.

$$i\Delta\mathcal{G}(\vec{p}) = i \frac{(\delta_\Phi + \delta_m)m^2 - p^2\delta_\Phi}{(p^2 - m^2)^2} - \frac{i\Sigma(\vec{p})}{(p^2 - m^2)^2} \quad (24)$$

$$\Delta\mathcal{G}(\vec{p}) = \frac{(\delta_\Phi + \delta_m)m^2 - p^2\delta_\Phi - \Sigma(\vec{p})}{(p^2 - m^2)^2} \quad (25)$$

6. THE Σ LOOP

$$i\Sigma(\vec{p}) = (-i\lambda) \frac{S}{(2\pi)^4} \int d^4q \frac{i}{q^2 - m^2 + i\epsilon} \quad (26)$$

S is the symmetry factor (Fig. 7).

$$S = \frac{4 \times 3}{4!} = \frac{1}{2} \quad (27)$$

$$\Sigma = -\frac{i\lambda}{2(2\pi)^4} \int \frac{d^4q}{q^2 - m^2 + i\epsilon} \quad (28)$$

6.1. UV Divergence

$$\Sigma \sim \int_0^\Lambda dq \frac{q^3}{q^2} = \int_0^\Lambda dq q \sim \Lambda^2, \quad \Lambda \rightarrow \infty \quad (29)$$

Σ has a UV divergence.

6.2. DimReg

$$\Sigma(d) = -\frac{i\mu^{4-d}\lambda}{2(2\pi)^d} \int \frac{d^d q}{q^2 - m^2 + i\epsilon} \quad (30)$$

From my Real Scalar Field Notes

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^b} = \frac{i(-1)^{-b}(2\pi)^d}{(4\pi)^{d/2}\Delta^{b-d/2}} \frac{\Gamma(b-d/2)}{\Gamma(b)}. \quad (31)$$

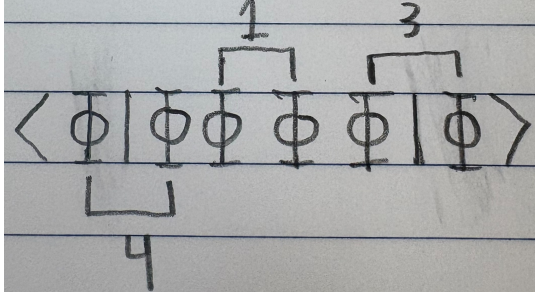


FIG. 7. Symmetry factor for Fig. 5.

$$\frac{1}{(2\pi)^d} \int \frac{d^d p}{p^2 - m^2 + i\epsilon} = \frac{\Gamma(1 - d/2)}{i(4\pi)^{d/2} m^{2-d}} \quad (32)$$

$$\Sigma(d) = -\frac{\mu^{4-d} \lambda \Gamma(1 - d/2)}{2(4\pi)^{d/2} m^{2-d}} \quad (33)$$

$$\Sigma = \Sigma(4) = \Sigma(4 - \epsilon), \quad \epsilon \rightarrow 0 \quad (34)$$

$$\Sigma = -\frac{\mu^\epsilon \lambda \Gamma(\epsilon/2 - 1)}{2(4\pi)^{2-\epsilon/2} m^{\epsilon-2}} \quad (35)$$

$$\Sigma = -\frac{\lambda m^2}{32\pi^2} \left(\frac{2\sqrt{\pi}\mu}{m} \right)^\epsilon \Gamma\left(\frac{\epsilon}{2} - 1\right) \quad (36)$$

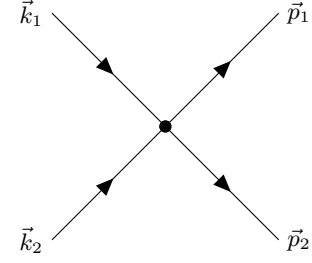
I take $\epsilon \rightarrow 0$.

$$\Sigma = \frac{\lambda m^2}{32\pi^2} \left[\frac{2}{\epsilon} + \ln\left(\frac{4\pi\mu^2}{m^2} e^{1-\gamma}\right) \right] \quad (37)$$

Here $\gamma \approx 0.577216$ is the Euler-Mascheroni constant.

7. MASS & FIELD COUNTERTERMS

$$\Delta\mathcal{G}(p^2) = \Delta\mathcal{G}(\vec{p}) = \frac{(\delta_\Phi + \delta_m)m^2 - p^2\delta_\Phi - \Sigma}{(p^2 - m^2)^2} \quad (38)$$


 FIG. 8. $\mathcal{C}$

$$\Sigma = \frac{\lambda m^2}{16\pi^2 \epsilon} + m^2 F, \quad F = \frac{\lambda}{32\pi^2} \ln\left(\frac{4\pi\mu^2}{m^2} e^{1-\gamma}\right) \quad (39)$$

F is the finite part.

$$(p^2 - m^2)^2 \Delta\mathcal{G}(p^2) = (\delta_\Phi + \delta_m)m^2 - p^2\delta_\Phi - \Sigma \quad (40)$$

$$(p^2 - m^2)^2 \Delta\mathcal{G}(p^2) = \left(\delta_\Phi + \delta_m - \frac{\lambda}{16\pi^2 \epsilon} - F \right) m^2 - p^2\delta_\Phi \quad (41)$$

I set the counterterms with minimal subtraction.

$$\boxed{\delta_\Phi = 0}, \quad \boxed{\delta_m = \frac{\lambda}{16\pi^2 \epsilon}} \quad (42)$$

$$(p^2 - m^2)^2 \Delta\mathcal{G}(p^2) = -m^2 F \quad (43)$$

$$\boxed{\left(\frac{p^2}{m^2} - 1 \right)^2 \Delta\mathcal{G}(p^2) = -\frac{\lambda}{32\pi^2 m^2} \ln\left(\frac{4\pi\mu^2}{m^2} e^{1-\gamma}\right)} \quad (44)$$

8. 2 TO 2 SCATTERING

I set δ_λ with the $\Phi\Phi \rightarrow \Phi\Phi$ S-matrix, not the 4-point Green's function. The second order matrix element is

$$\mathcal{M}^{(2)} = \mathcal{M}_0 + \Delta\mathcal{M}, \quad \mathcal{M}_0 = \mathcal{M}^{(1)}, \quad \Delta\mathcal{M} = \Delta\mathcal{M}^{(2)}. \quad (45)$$

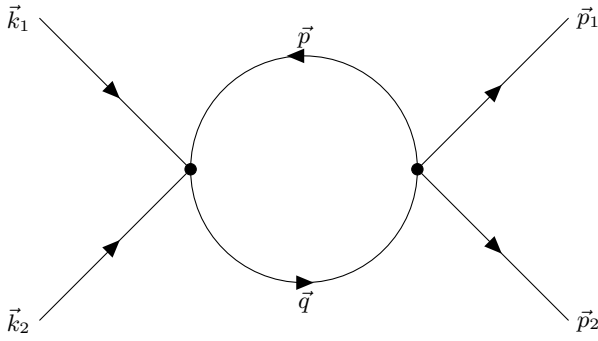
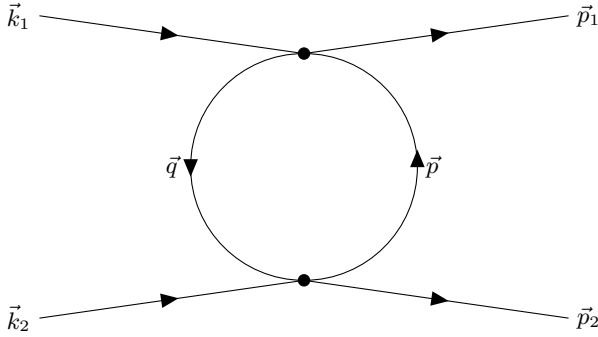
This time the leading term is first order.

8.1. Lowest Order

There is one leading order diagram ($\mathcal{C}$ in Fig. 8). I use Mandelstam variables.

$$\begin{aligned} s &= (\vec{k}_1 + \vec{k}_2)^2 = (\vec{p}_1 + \vec{p}_2)^2, \\ t &= (\vec{k}_1 - \vec{p}_1)^2 = (\vec{k}_2 - \vec{p}_2)^2, \\ u &= (\vec{k}_1 - \vec{p}_2)^2 = (\vec{k}_2 - \vec{p}_1)^2. \end{aligned} \quad (46)$$

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{k}_1 + \vec{k}_2 \quad (47)$$

FIG. 9. $\star[s]$ FIG. 10. $\star[t]$

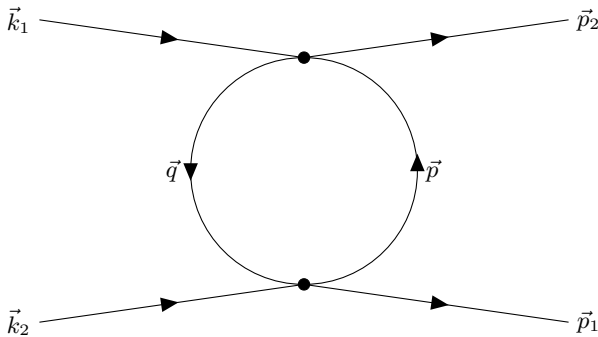
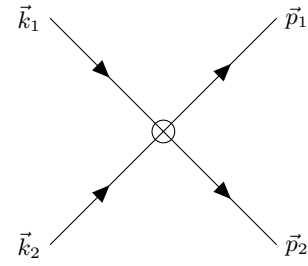
$$\mathfrak{C} = -i\lambda\delta^{(4)}(\vec{K} - \vec{P}) \quad (48)$$

$$\delta^{(4)}(\vec{K} - \vec{P})i\mathcal{M}_0 = \mathfrak{C} \quad (49)$$

$$\boxed{\mathcal{M}_0 = -\lambda} \quad (50)$$

8.2. Second Order

At second order there are 3 interaction diagrams (Figs. 9 - 11) and one counterterm diagram (Fig. 12).

FIG. 11. $\star[u]$ FIG. 12. $\star$

$$\delta^{(4)}(\vec{K} - \vec{P})i\Delta\mathcal{M} = \star[s] + \star[t] + \star[u] + \star \quad (51)$$

$$\star = -i(2\delta_\Phi + \delta_\lambda)\lambda\delta^{(4)}(\vec{K} - \vec{P}) \quad (52)$$

$$\delta_\Phi = 0 \quad (53)$$

$$\star = -i\delta_\lambda\lambda\delta^{(4)}(\vec{K} - \vec{P}) \quad (54)$$

$$\delta^{(4)}(\vec{K} - \vec{P})i\Delta\mathcal{M} = \star[s] + \star[t] + \star[u] - i\delta_\lambda\lambda\delta^{(4)}(\vec{K} - \vec{P}) \quad (55)$$

9. SOLAR DIAGRAMS

$$\star[s] = \frac{S(-i\lambda)^2}{(2\pi)^4} \int d^4q d^4p \frac{i}{p^2 - m^2 + i\epsilon} \frac{i}{q^2 - m^2 + i\epsilon} \delta^{(4)}(\vec{k}_1 + \vec{k}_2 + \vec{p} - \vec{q}) \delta^{(4)}(\vec{q} - \vec{p} - \vec{p}_1 - \vec{p}_2) \quad (56)$$

This is a one-loop diagram so it gets one $(2\pi)^{-4}$ factor. S is the symmetry factor. In my Real Scalar Field Notes I show $S = 1/2$ for this diagram (I won't repeat the calculation).

$$\star[s] = \frac{i^4\lambda^2}{2(2\pi)^4} \int d^4q d^4p \frac{\delta^{(4)}(\vec{p} - \vec{q} + \vec{K})\delta^{(4)}(\vec{q} - \vec{p} - \vec{P})}{(p^2 - m^2 + i\epsilon)(q^2 - m^2 + i\epsilon)} \quad (57)$$

$$\star[s] = \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int \frac{d^4p}{(p^2 - m^2 + i\epsilon)[(\vec{p} + \vec{P})^2 - m^2 + i\epsilon]} \quad (58)$$

9.1. Feynman Parameters

$$A = (\vec{p} + \vec{P})^2 - m^2 + i\epsilon, \quad B = p^2 - m^2 + i\epsilon \quad (59)$$

$$\star[s] = \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int \frac{d^4p}{AB} \quad (60)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[A + (B - A)x]^2} \quad (61)$$

$$\begin{aligned} \star[s] &= \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int d^4p \\ &\int_0^1 \frac{dx}{[p^2 + P^2 + 2\vec{p} \cdot \vec{P} - m^2 - (P^2 + 2\vec{p} \cdot \vec{P})x + i\epsilon]^2} \end{aligned} \quad (62)$$

$$s = (\vec{p}_1 + \vec{p}_2)^2 = P^2 \quad (63)$$

$$\begin{aligned} \star[s] &= \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int d^4p \\ &\int_0^1 \frac{dx}{[p^2 + s + 2\vec{p} \cdot \vec{P} - m^2 - (s + 2\vec{p} \cdot \vec{P})x + i\epsilon]^2} \end{aligned} \quad (64)$$

$$\begin{aligned} \star[s] &= \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int d^4p \\ &\int_0^1 \frac{dx}{[p^2 + 2(1-x)\vec{P} \cdot \vec{p} - m^2 - xs + s + i\epsilon]^2} \end{aligned} \quad (65)$$

9.2. Complete the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (66)$$

$$x \rightarrow \vec{p}, \quad a \rightarrow 1, \quad b \rightarrow 2(1-x)\vec{P}, \quad c \rightarrow s - m^2 - xs + i\epsilon \quad (67)$$

$$\begin{aligned} \star[s] &= \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int d^4p \int_0^1 \\ &\frac{dx}{\left[\vec{p} + (1-x)\vec{P} \right]^2 - m^2 - (1-x)^2 \vec{P}^2 + s - xs + i\epsilon}^2 \end{aligned} \quad (68)$$

$$\begin{aligned} \star[s] &= \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int d^4p \int_0^1 \\ &\frac{dx}{\left[\vec{p} + (1-x)\vec{P} \right]^2 - \Delta + i\epsilon}^2 \end{aligned} \quad (69)$$

$$\Delta = m^2 - (1-x)xs \quad (70)$$

9.3. Shift Integrand

$$\begin{aligned} \star[s] &= \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int_0^1 dx \\ &\int \frac{d^4p}{\left[\vec{p} + (1-x)\vec{P} \right]^2 - \Delta + i\epsilon}^2 \end{aligned} \quad (71)$$

$$\vec{q} = \vec{p} + (1-x)\vec{P}, \quad d^4q = d^4p \quad (72)$$

$$\star[s] = \frac{\lambda^2}{2(2\pi)^4} \delta^{(4)}(\vec{K} - \vec{P}) \int_0^1 dx \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (73)$$

9.4. UV Divergence

$$\star[s] \sim \int_1^\Lambda dq \frac{q^3}{q^4} = \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (74)$$

$\star[s]$ has a UV divergence.

9.5. DimReg

$$\star[s] = \frac{\delta^{(4)}(\vec{K} - \vec{P})}{2} \int_0^1 dx I \quad (75)$$

$$I = \frac{\lambda^2}{(2\pi)^4} \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (76)$$

$$I(d) = \frac{\mu^{2(4-d)} \lambda^2}{(2\pi)^d} \int \frac{d^d q}{(q^2 - \Delta + i\epsilon)^2} \quad (77)$$

In my Real Scalar Field Notes I derived

$$\int \frac{d^d p}{(p^2 - \Delta + i\epsilon)^2} = \frac{i\pi^{d/2}}{\Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right). \quad (78)$$

$$I(d) = \frac{i\mu^{2(4-d)} \lambda^2}{2^d \pi^{d/2} \Delta^{2-d/2}} \Gamma\left(\frac{4-d}{2}\right) \quad (79)$$

$$I(d) = \frac{i\mu^8 \lambda^2}{\Delta^2} \left(\frac{\sqrt{\Delta/\pi}}{2\mu^2} \right)^d \Gamma\left(\frac{4-d}{2}\right) \quad (80)$$

$$I = I(4) = I(4 - \epsilon), \quad \epsilon \rightarrow 0 \quad (81)$$

$$I = \frac{i\mu^8\lambda^2}{\Delta^2} \left(\frac{\sqrt{\Delta/\pi}}{2\mu^2} \right)^{4-\epsilon} \Gamma\left(\frac{\epsilon}{2}\right) \quad (82)$$

$$I = i \left(\frac{\lambda}{4\pi} \right)^2 \left(\frac{2\mu^2}{\sqrt{\Delta/\pi}} \right)^\epsilon \Gamma\left(\frac{\epsilon}{2}\right) \quad (83)$$

I take $\epsilon \rightarrow 0$.

$$I = i \left(\frac{\lambda}{4\pi} \right)^2 \left[\frac{2}{\epsilon} + \ln \left(\frac{4\pi\mu^4}{\Delta e^\gamma} \right) \right] \quad (84)$$

$$\star[s] = \frac{\delta^{(4)}(\vec{K} - \vec{P})}{2} \int_0^1 dx I \quad (85)$$

$$\star[s] = \left(\frac{\lambda}{4\pi} \right)^2 \frac{i\delta^{(4)}(\vec{K} - \vec{P})}{2} \int_0^1 dx \left[\frac{2}{\epsilon} + \ln \left(\frac{4\pi\mu^4}{\Delta e^\gamma} \right) \right] \quad (86)$$

$$\star[s] = \left(\frac{\lambda}{4\pi} \right)^2 \frac{i\delta^{(4)}(\vec{K} - \vec{P})}{2} \left[\frac{2}{\epsilon} \int_0^1 dx + \int_0^1 dx \ln \left(\frac{4\pi\mu^4}{\Delta e^\gamma} \right) \right] \quad (87)$$

$$\star[s] = \left(\frac{\lambda}{4\pi} \right)^2 \frac{i\delta^{(4)}(\vec{K} - \vec{P})}{2} \left[\frac{2}{\epsilon} + \int_0^1 dx \ln \left(\frac{4\pi\mu^4}{\Delta e^\gamma} \right) \right] \quad (88)$$

$$\star[s] = i\delta^{(4)}(\vec{K} - \vec{P}) \left[\left(\frac{\lambda}{4\pi} \right)^2 \frac{1}{\epsilon} + \mathcal{F}_s \right] \quad (89)$$

$$\mathcal{F}_s = \frac{1}{2} \left(\frac{\lambda}{4\pi} \right)^2 \int_0^1 dx \ln \left(\frac{4\pi\mu^4}{\Delta e^\gamma} \right) \quad (90)$$

$$\star[s] = i\delta^{(4)}(\vec{K} - \vec{P}) \left(\frac{\lambda^2}{16\pi^2\epsilon} + \mathcal{F}_s \right) \quad (91)$$

$\mathcal{F}_s$ is the finite part of $\star[s]$.

9.6. t & u Diagrams

In Eq. 91 only the finite part depends on s . Then by symmetry

$$\begin{aligned} \star[t] &= i\delta^{(4)}(\vec{K} - \vec{P}) \left(\frac{\lambda^2}{16\pi^2\epsilon} + \mathcal{F}_t \right), \\ \star[u] &= i\delta^{(4)}(\vec{K} - \vec{P}) \left(\frac{\lambda^2}{16\pi^2\epsilon} + \mathcal{F}_u \right). \end{aligned} \quad (92)$$

I do not need to know what the finite parts ($\mathcal{F}_t$ & $\mathcal{F}_u$) are.

10. COUPLING COUNTERTERM

$$\delta^{(4)}(\vec{K} - \vec{P}) i\Delta\mathcal{M} = \star - i\delta_\lambda \lambda \delta^{(4)}(\vec{K} - \vec{P}) \quad (93)$$

$$\star = \star[s] + \star[t] + \star[u] \quad (94)$$

$$\begin{aligned} \star[s] &= i\delta^{(4)}(\vec{K} - \vec{P}) \left(\frac{\lambda^2}{16\pi^2\epsilon} + \mathcal{F}_s \right) \\ \star[t] &= i\delta^{(4)}(\vec{K} - \vec{P}) \left(\frac{\lambda^2}{16\pi^2\epsilon} + \mathcal{F}_t \right) \\ \star[u] &= i\delta^{(4)}(\vec{K} - \vec{P}) \left(\frac{\lambda^2}{16\pi^2\epsilon} + \mathcal{F}_u \right) \end{aligned} \quad (95)$$

$$\star = i\delta^{(4)}(\vec{K} - \vec{P}) \left(\frac{3\lambda^2}{16\pi^2\epsilon} + \mathcal{F} \right) \quad (96)$$

$$\mathcal{F} = \mathcal{F}_s + \mathcal{F}_t + \mathcal{F}_u \quad (97)$$

$$\Delta\mathcal{M} = \frac{3\lambda^2}{16\pi^2\epsilon} - \delta_\lambda \lambda + \mathcal{F} \quad (98)$$

$$\boxed{\delta_\lambda = \frac{3\lambda}{16\pi^2\epsilon}} \quad (99)$$

$$\boxed{\Delta\mathcal{M} = \mathcal{F}} \quad (100)$$

Part II

2

First consider the integral (following Schwartz Sec. B.3.4)

$$I^{\alpha\beta} = \int d^d p p^\alpha p^\beta f(p^2). \quad (101)$$

Now comes a trick. This integral is a tensor so it must be proportional to the only available tensor with the correct symmetry: the Minkowski metric $\eta^{\alpha\beta}$. I find this argument hand wavy and non-intuitive. Is there really no other tensor around? But it is standard so I use it. Then

$$I^{\alpha\beta} = \kappa \eta^{\alpha\beta} \int d^d p p^2 f(p^2), \quad (102)$$

where p^2 is the only “natural choice” with the correct units and κ is a dimensionless constant of proportionality. Again I am very uncomfortable with these arguments. To find κ I contract Eqs. 101 & 102 with the metric.

$$\eta_{\alpha\beta} I^{\alpha\beta} = \int d^d p \eta_{\alpha\beta} p^\alpha p^\beta f(p^2) = \int d^d p p^2 f(p^2) \quad (103)$$

$$\eta_{\alpha\beta} I^{\alpha\beta} = \kappa \eta_{\alpha\beta} \eta^{\alpha\beta} \int d^d p p^2 f(p^2), \quad (104)$$

$$\eta_{\alpha\beta} \eta^{\alpha\beta} = \eta^\alpha_\alpha = \delta^\alpha_\alpha = d \quad (105)$$

$$\eta_{\alpha\beta} I^{\alpha\beta} = \kappa d \int d^d p p^2 f(p^2) \quad (106)$$

Plugging in Eq. 103 yields

$$\eta_{\alpha\beta} I^{\alpha\beta} = \kappa d \eta_{\alpha\beta} I^{\alpha\beta} \implies 1 = \kappa d \implies \kappa = 1/d. \quad (107)$$

$$I^{\alpha\beta} = \frac{\eta^{\alpha\beta}}{d} \int d^d p p^2 f(p^2) \quad (108)$$

$$\boxed{\int d^d p p^\alpha p^\beta f(p^2) = \frac{\eta^{\alpha\beta}}{d} \int d^d p p^2 f(p^2)} \quad (109)$$

Now consider

$$\mathbb{I}^{\alpha\beta\mu\nu} = \int d^d p p^\alpha p^\beta p^\mu p^\nu f(p^2). \quad (110)$$

Again only the metric tensor is available. Because it is even, $\eta_{\alpha\beta} = \eta_{\beta\alpha}$, the unique 4-index tensor I can construct with it is $\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu}$. Because of symmetry (no term is special) they all get the same constant of proportionality $\mathfrak{K}$.

$$\mathbb{I}^{\alpha\beta\mu\nu} = \mathfrak{K}(\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu}) \int d^d p p^4 f(p^2) \quad (111)$$

$$\boxed{p^4 = p^2 p^2} \quad (112)$$

Again I use contractions to solve for the constant.

$$\begin{aligned} \mathfrak{G} &= \eta_{\alpha\beta}\eta_{\mu\nu}\mathbb{I}^{\alpha\beta\mu\nu} = \int d^d p \eta_{\alpha\beta} p^\alpha p^\beta \eta_{\mu\nu} p^\mu p^\nu f(p^2) = \\ &\int d^d p p^2 p^2 f(p^2) = \int d^d p p^4 f(p^2) \end{aligned} \quad (113)$$

I plug this into Eq. 111.

$$\mathbb{I}^{\alpha\beta\mu\nu} = \mathfrak{K}(\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu})\mathfrak{G} \quad (114)$$

$$\eta_{\alpha\beta}\eta_{\mu\nu}\mathbb{I}^{\alpha\beta\mu\nu} = \eta_{\alpha\beta}\eta_{\mu\nu}\mathfrak{K}(\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu})\mathfrak{G} \quad (115)$$

$$\mathfrak{G} = \eta_{\alpha\beta}\eta_{\mu\nu}\mathfrak{K}(\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu})\mathfrak{G} \quad (116)$$

$$1 = \eta_{\alpha\beta}\eta_{\mu\nu}\mathfrak{K}(\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu}) \quad (117)$$

$$\frac{1}{\mathfrak{K}} = \eta_{\alpha\beta}\eta_{\mu\nu}\eta^{\alpha\beta}\eta^{\mu\nu} + \eta_{\alpha\beta}\eta_{\mu\nu}\eta^{\alpha\mu}\eta^{\beta\nu} + \eta_{\alpha\beta}\eta_{\mu\nu}\eta^{\alpha\nu}\eta^{\beta\mu} \quad (118)$$

$$\frac{1}{\mathfrak{K}} = \eta_{\alpha\beta}\eta^{\alpha\beta}\eta_{\mu\nu}\eta^{\mu\nu} + \eta_{\alpha\beta}\eta^{\alpha\mu}\eta_{\mu\nu}\eta^{\beta\nu} + \eta_{\alpha\beta}\eta^{\alpha\nu}\eta_{\mu\nu}\eta^{\beta\mu} \quad (119)$$

$$\frac{1}{\mathfrak{K}} = d^2 + \delta^\mu_\beta \delta^\beta_\mu + \delta^\nu_\beta \delta^\beta_\nu = d^2 + \delta^\beta_\beta + \delta^\beta_\beta = d^2 + 2\delta^\beta_\beta = d^2 + 2d \quad (120)$$

$$\frac{1}{\mathfrak{K}} = d^2 + 2d = d(d+2) \implies \mathfrak{K} = \frac{1}{d(d+2)} \quad (121)$$

$$\mathbb{I}^{\alpha\beta\mu\nu} = \frac{\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu}}{d(d+2)} \int d^d p p^4 f(p^2) \quad (122)$$

$$\begin{aligned} \int d^d p p^\alpha p^\beta p^\mu p^\nu f(p^2) = \\ \frac{\eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\mu}\eta^{\beta\nu} + \eta^{\alpha\nu}\eta^{\beta\mu}}{d(d+2)} \int d^d p p^4 f(p^2) \end{aligned} \quad (123)$$